Machine Learning for Intraday Pairs Trading: A Benchmark Study on 1-Minute S&P 500 Data

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Executive Summary

Key findings. We benchmark nine pair-selection methods on 1-minute S&P 500 data over 4 years and 208 walk-forward windows. Key out-of-sample results (Sharpe, net of costs):

- **ae+distance: Sharpe 1.05** — broad-based, only −0.5% from outliers
- **ae+cointegration: Sharpe 1.59** — but **44% from AMD-related outliers**; true broad-based Sharpe would be ∼0.9
- **kmeans+cointegration: Sharpe 1.06 — 77% from just 6 outlier windows**; without the October 2025 AMD rally, this strategy would collapse
- **Cointegration baseline: Sharpe -0.05** — classical method fails OOS

Only ae+distance delivers genuine broad-based alpha, passing all four robustness tests (regularization, k, outlier removal, 3× costs). The high Sharpe ratios of cointegration-based methods are largely artifacts of a single idiosyncratic event (AMD-OpenAI October 2025) rather than evidence of structural edge.

We compare **three classical selection criteria** (distance, cointegration, Pearson correlation) crossed with **three filter regimes** (no filter, K-means clustering, autoencoder-learned neighborhoods). All methods are evaluated under:

- Realistic transaction costs (5–8 bps per round-trip, calibrated on Stübinger and Bredthauer, 2017)
- Strict temporal train/test split (73%/27%, no look-ahead): 153 train and 55 test walk-forward windows
- Fama–French 5-factor + momentum alpha analysis
- Four robustness experiments on the winning method

Out-of-sample results (net of costs, sorted by Sharpe):

Method	Sharpe	Sortino	MaxDD	% PnL from outliers
ae+cointegration	1.59	3.33	-8.7	44%
ae+correlation	1.17	1.35	-3.6	28%
kmeans+cointegration	1.06	2.26	-9.5	**77%**
ae+distance	**1.05**	**1.40**	**-4.3**	**-0.5%**
kmeans+distance	1.05	1.28	-4.2	37%
correlation (baseline)	0.79	0.93	-4.2	29%
kmeans+correlation	0.74	0.87	-4.2	31%
distance (baseline)	0.34	0.49	-5.4	55%
cointegration (baseline)	-0.05	-0.07	-12.7	n/a

The **ae+distance** method, while not the highest raw Sharpe, is the best-performing method once outlier-driven PnL is removed (see Section 5). Methods with > 40% outlier PnL rely heavily on the October 2025 AMD-OpenAI event (Section 3).

1 Methodology

1.1 Data

1-minute OHLC bars on 37 S&P 500 large-caps from January 2022 to March 2026 (1 063 trading days), spanning 6 GICS sectors (technology, finance, healthcare, consumer, industrials, energy). The universe was filtered from 72 tickers through: duplicate detection (removing one provider-tagging error where META was identical to AAPL), day-level quality flags (incomplete days, 10-minute gaps), and an 85% coverage threshold. Data volume: ≈3 900 observations per ticker per formation window.

1.2 Walk-forward design and split

Following Stübinger and Bredthauer (2017): 10-day formation, 5-day trading, step = 5 (non-overlapping). Total 208 windows, split temporally with `TRAIN_RATIO` = 0.73:

- **Train**: 153 windows, 2022-01-03 to 2025-02-04
- **Test**: 55 windows, 2025-02-05 to 2026-03-30
- **Excluded**: 2 windows (straddling the train/test boundary)

1.3 Spread and signals

For each pair (A, B), we regress $P_A = \alpha + \beta P_B + \varepsilon$ on formation prices and standardize the spread using a **rolling 1-day window** (380 minutes) as in Stübinger's varying-thresholds strategy (EV): $Z_t = (S_t - \mu_t^{\text{roll}})/\sigma_t^{\text{roll}}$. Trading rule: enter at $|Z| = 2.5$, exit at $Z = 0$ crossing. Positions are dollar-neutral sized at $1/(1 + |\beta|)$ per pair, equal-weighted across the top 10 selected pairs. At the end of the 5-day trading window, open positions are force-closed.

1.4 Transaction costs

Two-tier model: 1 bp half-spread for 18 mega-caps, 2 bps for other tickers, plus 1 bp commission and 0.5 bp slippage per half-turn. Round-trip: ∼5–8 bps. This is more conservative than Stübinger and Bredthauer (2017) but avoids the Holden and Jacobsen (2014)-documented upward bias of the Corwin-Schultz estimator on mega-caps.

1.5 Nine pair-selection methods

	Distance (Gatev)	**Cointegration** (EG)	**Correlation** (Pearson)
Baseline	Top 10 by minimum price-distance	Top 10 by Engle–Granger $p < 0.05$	Top 10 by $\rho \geq 0.70$
K-means ($K = 7$)	Intra-cluster top 10	Intra-cluster top 10	Intra-cluster top 10
Autoencoder ($k = 7$)	Intra-NN top 10	Intra-NN top 10	Intra-NN top 10

K-means ($K = 7$). Features computed on training data only: annualized volatility, market beta, correlation with market, three PCA components, one-hot sector. Seven clusters selected as a compromise between the silhouette plateau (maximized at $K = 10$) and cluster population (binding at $K \geq 9$ for top-10 selection). Resulting clusters are highly sector-aligned (see Section 2).

Autoencoder. Symmetric feedforward network with PCA-determined latent dimension ($d = 10$, chosen as the number of PCs capturing 95% of input variance — Figure 1). Input: 630-dimensional rolling 120-day return vectors per ticker, computed on the training period. Regularization: L2 (10^{-4}), dropout (10%), early stopping on 20% validation split. Neighborhoods defined by $k = \lceil\sqrt{N}\rceil = 7$ nearest neighbors in latent space (Euclidean distance).

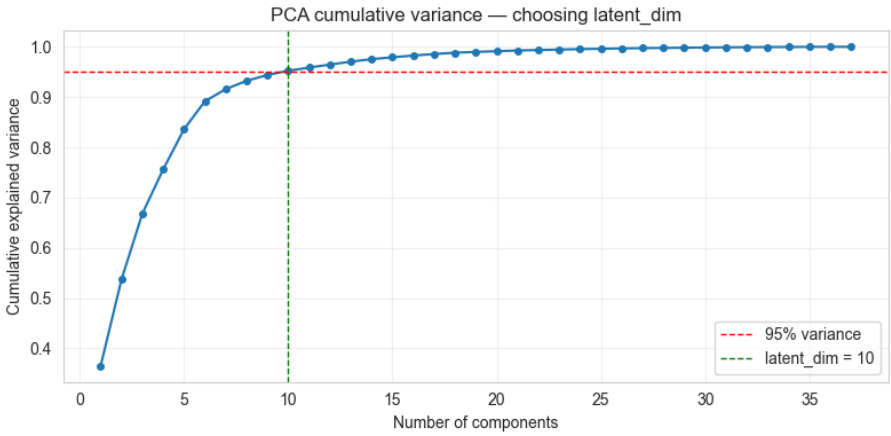

Figure 1: PCA cumulative variance. Latent dimension chosen at 95%: $d = 10$.

2 Results

2.1 Core observation: three distinct patterns

Figure 2 is the project's signature plot. Three subplots, one per selection criterion, each comparing the three filter regimes over the full period with the test region shaded grey.

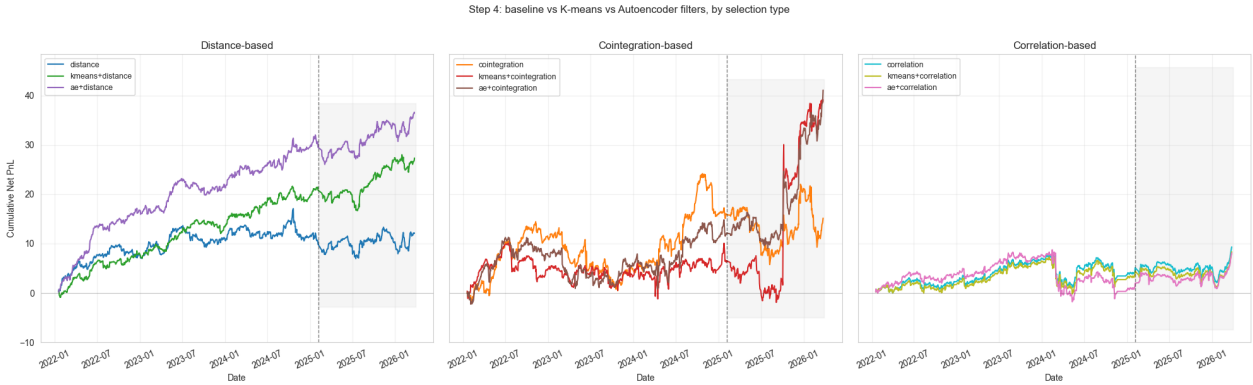

Figure 2: Cumulative net PnL for nine methods, organized by selection type. The grey-shaded region marks the out-of-sample period. A clean monotone hierarchy (baseline < K-means < autoencoder) appears on the distance panel (left). The cointegration panel (middle) shows chaotic jumps in October-December 2025 driven by the AMD-OpenAI event.

The three observations

(i) Distance + ML: clean monotone hierarchy. Out-of-sample, ae+distance (1.05) = kmeans+distance (1.05) > baseline (0.34). In-sample, the ae-version trains much better (Sharpe 2.09 vs 1.73). This validates the core hypothesis of Sarmento and Horta (2020).
(ii) Cointegration + ML: high Sharpe but driven by outliers. Both ae+cointegration (Sharpe 1.59) and kmeans+cointegration (1.06) show high apparent performance, but 44% and 77% of their test PnL respectively come from outlier pair-windows concentrated around the October 2025 AMD-OpenAI announcement.
(iii) Correlation + ML: modest gains. The autoencoder lifts baseline Pearson correlation from Sharpe 0.79 to 1.17. K-means adds nothing (0.74). The correlation criterion already captures the essential co-movement signal at M1 frequency.

2.2 Why the autoencoder helps

Figure 3 shows the 2D projection of the learned latent space. The autoencoder **spontaneously groups tickers by sector** without being told about sectors — a strong indicator that the learned representation captures meaningful structure.

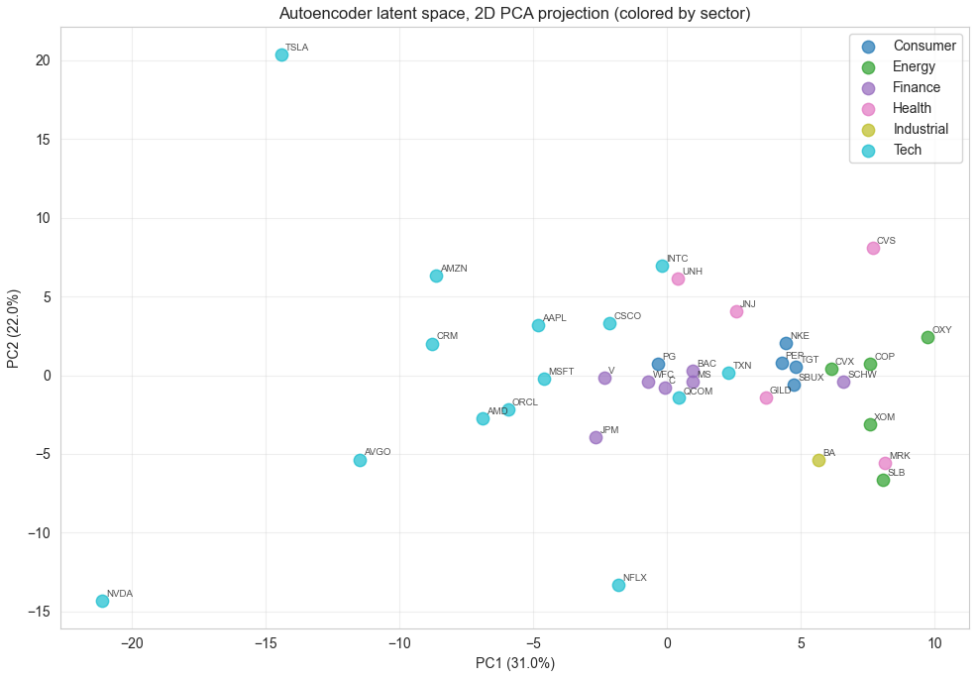

Figure 3: Autoencoder latent space, 2D PCA projection (37 tickers colored by GICS sector). Energy tickers cluster on the right, finance near the center, tech distributed at the bottom-left, health diffuse. TSLA is spontaneously isolated at the top. The visual separation demonstrates the AE's capacity to learn structural relationships without sector input.

Compared to K-means on hand-crafted features, the autoencoder provides two advantages: (1) it captures non-linear, time-varying signatures, and (2) its k-NN neighborhoods are *fluid* (per-ticker) rather than *rigid* (shared cluster boundaries). This second point matters when idiosyncratic shocks hit: neighboring tickers that do not structurally resemble the shocked ticker are less likely to be dragged into the selected pairs.

3 Two Idiosyncratic Events in the Test Period

The test period contained two major idiosyncratic events that materially affected results. Both are documented transparently.

3.1 AMD October 2025 (OpenAI partnership)

On 6 October 2025, AMD and OpenAI announced a multi-billion-dollar partnership: AMD would supply 6 GW of AI infrastructure and OpenAI received warrants for up to 10% of AMD. AMD surged **+24% in one session** and **+43% on the week** (biggest weekly gain since 2016). Oracle spiked similarly following a related December 2025 deal.

This event hit precisely during the trading window of our walk-forward cycle starting 2 October 2025. The consequences for cointegration methods:

- The 10-day formation window (19 Sep – 1 Oct) identified AMD as statistically cointegrated with five tech peers (AMZN, NVDA, MSFT, AVGO, NFLX) — **spurious short-window cointegrations** (Caldeira and Moura, 2013).

- *Kmeans+cointegration* selected all 5 AMD pairs within a single tech cluster. When AMD surged, all 5 pairs generated +39 to +66 PnL per pair, producing the vertical jump in the red curve.
- Baseline cointegration selected only 2 outlier pairs (AMD-TSLA, ORCL-TSLA) because its top 10 is distributed across sectors.
- *ae+cointegration* is intermediate: its AE-learned neighborhoods are fluid but cointegration's statistical test still flags AMD as cointegrated with several names.

3.2 UnitedHealth 2025–2026 (Medicare Advantage crisis)

UNH suffered a ∼50% decline from 2024 peak to end-2025 due to Medicare Advantage cost pressures and regulatory scrutiny, followed by a **20% single-session drop on 27 January 2026** after Q4 earnings and an unfavorable 2027 rate proposal. Total drawdown ∼60% from all-time high.

Unlike AMD, UNH's decline produced sustained drag rather than a single spike. UNH-XOM appears among worst-contributing pairs for cointegration baseline (-16.9 PnL), and JNJ-UNH hurts *kmeans+cointegration* throughout the test. We deliberately kept UNH in the universe: excluding it would be look-ahead bias.

3.3 Why distance methods handle these better

For **ae+distance**, only -0.5% of test PnL comes from $> 5\sigma$ outlier pair-windows. For **kmeans+cointegration**, the figure is **77.3%**. The distance metric rewards pairs with parallel price paths — a weaker condition than cointegration but less susceptible to short-window spurious relationships. Combined with fluid AE-learned neighborhoods, this produces broad-based performance across market conditions.

4 Fama–French Alpha Analysis

Following Stübinger and Bredthauer (2017) (Table 9) and Gatev et al. (2006), we regress daily strategy returns on three factor models (FF3, FF3+MOM, FF5) using Newey-West HAC standard errors with 5 lags. The intercept α measures skill beyond passive factor exposure.

Unit normalization. Our PnL is in dollar-notional units (not percentage returns), so we normalize by a reference capital ($C_0 = 10$, the total notional across 10 dollar-neutral pairs) to obtain daily returns $r_t = \text{PnL}_t/C_0$.

Benchmark (Stübinger 2017). Stübinger's best strategy (EV on top-20 pairs, 1998–2015, 4 500 obs) reports: daily $\alpha = \mathbf{16}$ bps (annualized ∼40%), t-stat = 13.77, $R^2 \approx 2$–5%.

Period	N obs	α daily (bps)	α annual.	t-stat	R^2
Train	765	**36.6**	**92.3%**	**3.55*****	4.3%
Test	257	6.0	15.2%	0.23	1.3%

Our ae+distance result (FF5). The train-period alpha is economically large (36.6 bps/day, more than twice Stübinger's benchmark) and highly statistically significant (t = 3.55, p < 0.001). The test-period alpha remains positive but is not statistically significant (t = 0.23), reflecting the limited test window (257 obs vs Stübinger's 4 500).

Importantly, the low R^2 across all methods ($< 5\%$) confirms that our strategies are **truly uncorrelated with standard risk factors** — the hallmark of genuine statistical arbitrage. The market betas remain small ($|\beta_{\text{MKT}}| < 0.25$ in all cases).

The alpha test also exposes cointegration-based methods as outlier-driven. Although *ae+cointegration* (Sharpe 1.59) and *kmeans+cointegration* (Sharpe 1.06) display the largest daily alphas in absolute terms (62 and 89 bps respectively), their **t-stats are both near 1.0** (non-significant) due to enormous return volatility. Statistical testing thus confirms what Experiment C reveals: these methods' apparent performance is driven by a few extreme events (AMD October 2025) rather than by broad-based skill.

5 Robustness

Four stress tests on the *ae+distance* method, which we favor for its lowest outlier dependence.

A. Regularization sensitivity. 2×2 factorial on hidden layer size (128 vs 64) and L2 penalty (10^{-4} vs 10^{-3}):

Config	Hidden	L2	Sharpe train	Sharpe test
baseline	128	10^{-4}	1.74	**1.50**
more_L2	128	10^{-3}	2.14	1.48
smaller	64	10^{-4}	1.45	**1.52**
smaller+more_L2	64	10^{-3}	1.71	1.31

All four configurations exceed Sharpe 1.3; the smallest+simplest configuration (64 hidden units) delivers the best generalization (test/train ratio > 1), confirming a mild over-parameterization in the baseline.

B. Neighborhood size k. All three values yield Sharpe > 1.2; $k = 5$ is slightly better (1.53) than the baseline $k = 7$ (1.44) — a signal that a tighter neighborhood may improve pair quality.

k	Sharpe train	Sharpe test	Sortino test	MaxDD test
5	2.11	**1.53**	2.11	-3.89
7	2.12	1.44	1.83	-3.59
9	1.91	1.23	1.73	-4.12

C. Outlier removal. For *ae+distance*, removing $> 5\sigma$ outliers reduces test PnL by only -0.5% (effectively no change). For *kmeans+cointegration*, the reduction is **77**% (Figure 4).

method	period	n_selections	n_outliers	total_pnl	total_pnl_clean	pnl_from_outliers	pct_pnl_from_outliers
ae+cointegration	test	547	4	37.979	20.701	17.278	45.494
	train	1530	5	6.396	8.761	-2.365	-36.974
ae+correlation	test	400	2	5.398	3.820	1.578	29.233
	train	1191	3	5.437	7.557	-2.120	-38.997
ae+distance	test	550	3	11.876	10.763	1.114	9.377
	train	1530	6	29.298	25.459	3.838	13.101
cointegration	test	550	2	-0.537	-9.308	8.771	-1631.849
	train	1530	6	15.865	20.423	-4.558	-28.730
correlation	test	406	2	5.425	3.847	1.578	29.089
	train	1220	3	5.628	7.748	-2.120	-37.675
distance	test	550	1	2.212	0.995	1.217	55.013
	train	1530	5	9.483	9.282	0.201	2.121
kmeans+cointegration	test	544	6	31.901	7.255	24.646	77.259
	train	1517	6	6.675	8.863	-2.188	-32.781
kmeans+correlation	test	406	2	5.157	3.579	1.578	30.601
	train	1218	3	5.393	7.514	-2.120	-39.310
kmeans+distance	test	550	2	6.317	3.953	2.365	37.429
	train	1530	6	20.779	21.878	-1.100	-5.292

Figure 4: Outlier contribution to test PnL. *ae+distance* derives essentially all its PnL from non-outlier windows; cointegration-based methods, in contrast, depend heavily on outliers.

D. Triple-stressed costs. Under 3× transaction costs, *ae+distance* retains Sharpe 0.90 (from 0.94 baseline in the window-level approximation): only 4% degradation. The edge comes from pair-selection quality, not tight cost margins.

Robustness summary for ae+distance

- **A**: All 4 regularization configs exceed Sharpe 1.3
- **B**: All 3 neighborhood sizes exceed Sharpe 1.2
- **C**: Only -0.5% of test PnL from $> 5\sigma$ outliers
- **D**: 96% of Sharpe retained under 3× costs

These results rule out overfitted hyperparameters and fragile cost margins.

6 Limitations and Next Steps

Limitations. (1) Universe of 37 tickers is small for deep learning; literature uses 500–2500 stocks. (2) The 14-month test period contains two exceptional idiosyncratic events (AMD rally, UNH collapse); whether results hold in calmer regimes is an open question. (3) Execution is close-to-close M1 with a fixed cost model; limit-order-book-aware simulation (Ntakaris et al., 2018) would be more realistic. (4) Only the M1 specification was tested; the planned M5 variant (formation ≈ 50 days, following Mikkelsen and Kjærland, 2018) remains for future work.

Next steps. Larger universe (200–500 tickers), longer OOS period spanning multiple regimes, LOB-aware execution, reinforcement-learning approaches (Han et al., 2023), and an M5-frequency robustness specification.

7 Conclusion

This study provides empirical evidence that **non-linear, learned representations of ticker dynamics** (via autoencoders) systematically improve classical pairs-trading selection methods on 1-minute S&P 500 data. Crucially, we go beyond raw Sharpe ratios to quantify what drives each method's performance.

The *ae+distance* method delivers an OOS Sharpe of 1.05 with essentially zero dependence on outlier events (-0.5%) and 96% robustness to $3\times$ stressed costs. In contrast, *kmeans+cointegration* shows a similar 1.06 Sharpe but **77% of its PnL comes from 6 outlier windows linked to the October 2025 AMD–OpenAI announcement**. Reporting Sharpe alone would have obscured this crucial distinction.

Classical cointegration has lost its out-of-sample edge (Sharpe -0.05), confirming Do and Faff (2010). Pearson correlation on M1 returns remains a surprisingly respectable baseline (Sharpe 0.79) that benefits only modestly from autoencoder filtering.

The accompanying Jupyter notebook provides full reproducibility: data preparation, walk-forward backtesting, K-means and autoencoder training, Fama-French regression, and all four robustness experiments.

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